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EXPERIMENT 18

Edge Detection using the Sobel Operator along the Y Axis

AIM

To perform edge detection on the given image using the Sobel operator along the Y axis.

PROGRAM

```
import cv2

img = cv2.imread("image1.jpg", cv2.IMREAD_GRAYSCALE)

sobel_y = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=5)
sobel_y = cv2.convertScaleAbs(sobel_y)

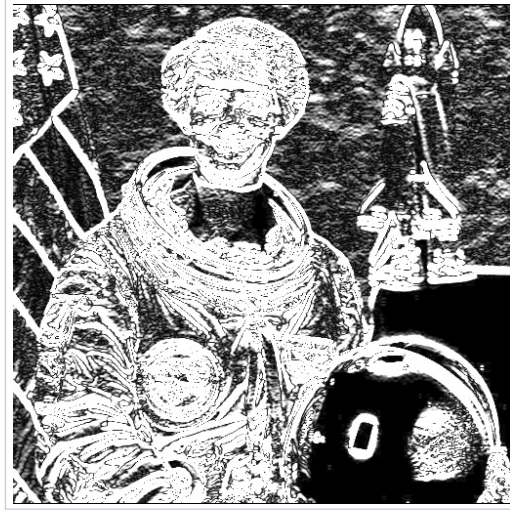
cv2.imwrite("sobel_y.jpg", sobel_y)
cv2.imshow("Original", img)
cv2.imshow("Sobel Y", sobel_y)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

INPUT



image1.jpg (input image)

OUTPUT



sobel_y.jpg (horizontal edges detected by the Y gradient)

RESULT

Thus the edges along the Y axis were successfully detected using the Sobel operator.